VERMONT
DEPARTMENT OF HEALTH

Vermont Community Resilience Index: Introduction and Methodology

October 2023

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Abstract

The Vermont Community Resilience Index (VT CRI) is an interactive health disparities tool. This tool aids public official officials examine community resilience among populations throughout Vermont. Community resilience is the capacity of a population to mitigate and recover from a public health crisis and personal health emergency in a manner that sustains physical, social, and mental health. The VT CRI tool consists of three components: composite index; indicator displays; population reference maps. The composite index integrates 11 socioeconomic and housing indicators into a single value assigned to each Census tract and municipality. There is a unique composite index foreach geographic frame, including statewide and each specific VDH district, hospital service area, and county. The community resilience composite index serves as a model to explore the social conditions which influence a community's ability to efficiently recover from a health crisis.

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Introduction

The Vermont Community Resilience Index (VT CRI) is a multifaceted tool that seeks to inform public health officials on the distribution of community resilience across the state. Resilience is a term that has numerous definitions across disciplines, but it is generally considered the ability of a subject to respond to, mitigate, and effectively recover from stressors with minimal assistance from external sources (Manyena, 2006). There is a wide variety of common subjects of a resilience framework including individual persons, societies, and whole environmental systems (Glonti et al., 2015; Holling, 1973; Masten et al., 2009). Similar in its large variety, example stressors of these subjects include infectious disease, mental health status, large-scale natural disasters, and environmental change (Cutter et al., 2010; Davydov et al., 2010; Nelson et al., 2007; Rutter, 1993).

In a public health domain, community resilience means the numerous expressions of on-the-ground daily reality that promote physical, social, and mental well-being (World Health Organization; Wulff et al., 2015). This resilience enables people to build vitality and the adaptability necessary to recover from a health crisis (Wulff et al., 2015). Informed by this understanding, the VT CRI views community resilience as the socioeconomic and housing capacity of a population to mitigate and recover from a public health crisis and personal health emergencies in a manner that sustains physical, social, and mental health.

The final form of the VT CRI is a public-facing application hosted on ArcGIS Online Experience Builder with three main components: composite index; indicator displays; population reference maps. The primary component of the tool is the composite index. This composite index combines multiple input indicators related to community resilience into a single geometric mean value. Each input indicator in the VT CRI is an accumulation of population variables derived from the Census Bureau's American Community Survey (ACS) 5-year estimate datasets. Secondly, the indicator display component examines the data of each input indicator through the statistic standard mean difference. Accompanying both the composite index and indicator displays, a series of population reference maps provide further contextual information about the population of the state but are not themselves indicators used in the calculation of the composite index. The data for each of these three components are presented at the Census tract and municipality geography units.

What is a Composite Index?

A composite index is a highly distilled output that attempts to measure the overarching trends among large macro processes (Nardo et al., 2005). Examples of large macro processes include resilience, environmental change, and economic conditions. A composite index draws together numerous measurements that relate to the specific macro process and combines them together to create a single value (OECD Committee on Statistics, 2004). Composite indices are

useful because they enable a user to measure the performance of a macro process across time and across geographies (OECD et al., 2008).

Key Definitions

- Community resilience - the capacity of a population to mitigate and recover from a public health crisis and personal health emergency in a manner that sustains physical, social, and mental health.
- Composite index - the compilation of individual indicators into a single metric to measure a multi-dimensional concept (OECD Committee on Statistics, 2004).
- Indicator – the data measures compiled together that create a composite index.
- Geography unit – a spatial unit of measurement. In the VT CRI, this includes Census tracts and municipalities.
- Census tract - small, relatively permanent statistical subdivision of a county or statistically equivalent entity that generally has a population size between 1,200 and 8,000 people, with an optimum size of 4,000 people (US Census Bureau, 2022).
- Municipality – equivalent to county subdivision unit of US Census Bureau and are most familiarly known as towns and cities.
- Area frame – an area of focus that determines the geographic extent to measure and present geography units. In the VT CRI, this includes the entire state and specific VDH districts, hospital service areas, and counties.
- Percentile rank - the proportion of values in a distribution that a specific value is greater than or equal to (Flanagan et al., 2011).
- VDH district – the 12 Vermont Department of Health local administrative units which carry out statewide public health initiatives and services at the local level.
- Hospital service area – the 13 Vermont local health care markets for hospital care measured by collections of zip codes whose residents receive most of their hospitalizations from the hospitals in that area (Law Insider).
- Theme – the thematic grouping of indicators. The Overall Theme represents all indicators of a selected year. The Socioeconomic Theme and Housing Theme represent two subsets of indicators of a selected year.

Indicators

The Vermont Community Resilience Index (VT CRI) draws together 11 population measures about the distribution of social and economic conditions across the state. These indicators were identified based upon their thematic relation to the community resilience definition. The selection of these indicators follows the precedent of considering social and economic indicators for public health purposes established by the CDC Social Vulnerability Index (CDC

SVI). In fact, of the 11 indicators integrated into the VT CRI, the indicators of the CDC SVI influenced 8 of the VT CRI indicators in some capacity.

The composite index contains 3 different themes. Themes are the assortment of indicators used in the calculation of the composite index. The primary theme is the Overall Theme which represents the combination of all available indicators. In addition to the Overall Theme, the 11 indicators are grouped into two subsets: Socioeconomic Theme and Housing Theme. For these two subsets, the composite index reflects only the indicators of that theme. To clarify, the values of the two subset themes do not influence the Overall Theme. Instead, the two subset themes provide the opportunity to review the composite index as a smaller assortment of indicators which share a thematic overlap.

Indicator Definitions

- Socioeconomic Theme – this theme reflects aspects of economic and material conditions of the population.
    1. Supportive Income – the percentage of the population with income greater than 200% of the Federal poverty level.
    2. Employment - the percentage of the adult population currently employed.
    3. Without Public Assistance – the percentage of households that have not received public assistance in the past 12 months.
    4. Median Household Income – median household income adjusted for inflation.
    5. High School Education – the percentage of adult population with a completed high school degree.
    6. Health Insurance – the percentage of the population with health insurance.
    7. Manageable Housing Costs – the percentage of housing units with housing costs less than 30% of household income.
- Housing Theme – this theme reflects aspects of housing quality and household composition of the population.
    8. Broadband Internet Subscription – the percentage of the population with access to a computer and broadband internet.
    9. Owner-Occupied Homes – the percentage of owner-occupied housing units.
    10. English Proficiency – the percentage of the population that speaks English “well” or better.
    11. Vehicle Access – the percentage of housing units with a vehicle available.

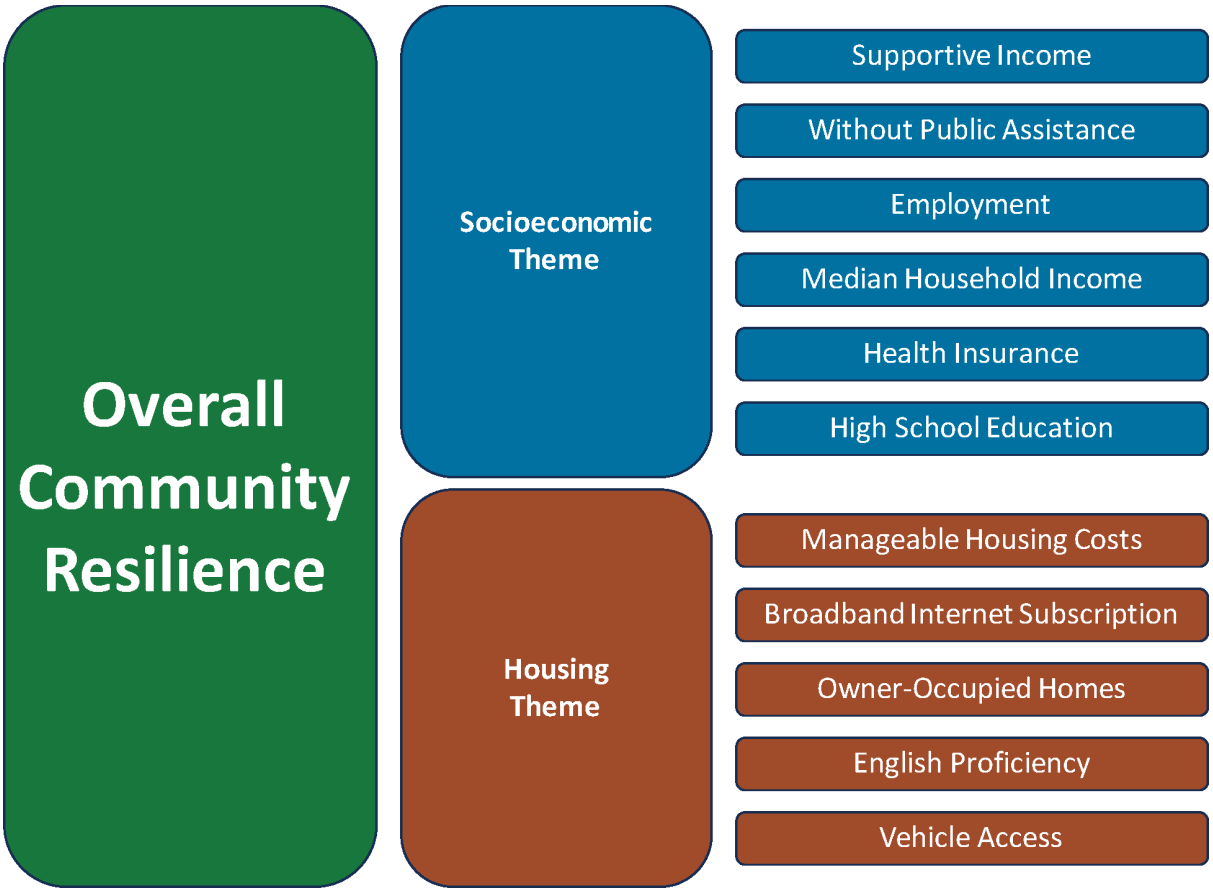


Figure 1. A visual representation of the 11 indicators and how they are thematically grouped by the Socioeconomic Theme and Housing Theme. Further, the visual representation demonstrates that the two themes are subsets of the overall Community Resilience Index.

Indicator Weighting

Every index requires the combination of different indicators. Combining indicators requires a decision of how to weigh them. Intentional indicator weighting determines the relative influence the value of an indicator has on the final composite index. As the 'Index Construction Steps' section of this document demonstrates, all indicators of the CRI are equally weighted. This means that all 11 indicators of the CRI have the same potential to influence the final value of the composite index. In other words, no indicator is purposefully weighted to have more influence than other indicators. Equal weighting is the most common weighting method used to aggregate indicator values in a composite index (Beccari, 2016).

Index Construction Steps

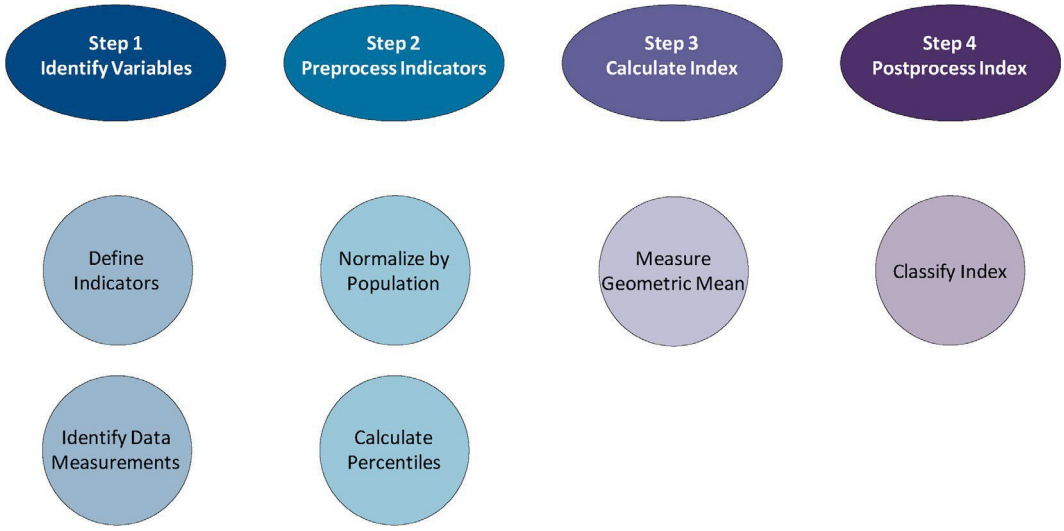


Figure 2. A summarized visual representation of the 4 steps to construct the composite index.

Step 1 – Identify Variables

First, we conducted a thorough review of local, state, and national social disparity indices throughout the United States. Among these indices, the CDC Social Vulnerability Index is the most prominent. We then evaluated the range of indicators used by these indices for their potential ability to contribute to a meaningful aspect of community resilience in the context of Vermont. We identified a total of 11 indicators from these sources to serve as inspiration for the construction of the VT CRI indicators.

With a general understanding of our 11 indicators, we identified plausible data measurements of US Census Bureau American Community Survey (ACS) 5-year estimates data tables that overlapped with the indicators. We considered the full extent of available measurements that aligned with our general indicator ideas, and then determined which to specifically use for the indicators. Through this process, we simultaneously identified the required ACS data measurements and created precise definitions for our indicators. For nearly all the 11 indicators, our final definitions are different from the definitions used by the indices we reviewed. These adjustments reflect the changes necessary to ensure that the indicators better capture the objectives of the VT Community Resilience Index. You can review these definitions in the 'Indicators' section of this document.

We then used a US Census Bureau API to pull the ACS data variables into a python script. We acquired the data of the variables for all Census tracts and municipalities in the state. Additionally, we acquired the data across multiple years, starting with 2008-2012 5-year

estimate data and completing at the most recent 2018-2022 5-year estimate data. Refer to the 'Data Appendix' section of this document to review the specific ACS variables used in the construction of the indicators and the available years for each indicator.

Step 2 – Preprocess Indicators

In the python script, we aggregated the numerous ACS variables to create a single value for each indicator. The values are aggregated simultaneously for each individual geography unit. These aggregated values are in the form of a fraction. The numerator of the fraction represents the total number of people measured by the indicator's definition per geography unit. The denominator of the fraction is the total population of the geography unit. It's important to recognize that the variables supporting each indicator are acquired from different ACS tables, which in turn are based on different original survey questions. As such, not all survey questions will contain the same response rates. This means that the reported total population of a given geography unit may vary across different ACS tables, and thus between different indicators. To compare the values of indicators within the same geography unit, we normalized the fraction indicator values. To do this, we divided the aggregated sum numerator by the total population denominator.

This normalized indicator value represents the proportion of a geography unit's population represented by the defined indicator. At this point, each geography unit has 11 normalized values assigned to it – one foreach indicator. Based on these normalized values, we calculated the percentile rank values of each indicator. The percentile rank output assigns a value of 0-100 to each geography unit and for each indicator. A percentile value of 0 signifies that the given geography unit has a lower normalized indicator value than all other geography units. A percentile value of 100 signifies that the given geography unit has a higher normalized indicator value than all other geography units.

We calculated percentile rank of indicators numerous times in different manners. Firstly, we processed the normalized indicator values separately for Census tract and municipality geography units. Secondly, we calculated the percentile rank at each area frame. For Census tracts, this means we measured the percentile rank of normalized indicators at a statewide frame and at each of Vermont's 14 counties. For municipalities, this means we measured the percentile rank of normalized indicators at a statewide frame and at each of Vermont's 12 VDH districts, 13 hospital service areas, and 14 counties. The ability to calculate a different percentile rank for each area frame reflects that percentile rank is a relative calculation. This means that the percentile value calculated from a geography unit's normalized indicator depends on what other geography units are included in the calculation. Since each area frame is a unique combination of geography units, each corresponding percentile rank is also unique.

Step 3 - Calculate Index

To construct the composite index, we used the geometric mean method to combine the 11 percentile rank values at each geography unit. We first multiplied the percentile rank values of each indicator by each other. Then, we acquired the *n*th root of that product. The *n* represents the number of values used in the calculation of the product, which for the Overall Theme is 11. This root value is the final Overall Theme of the composite index. We repeated this geometric mean method for the Socioeconomic Theme and Housing Theme subsets of indicators. Additionally, just like the percentile ranks from the Preprocess Indicators step, we repeated the geometric mean calculation at each area frame.

The final output of the geometric mean is a scale of 0-100. A geography unit with a value of 0 indicates that it has the lowest community resilience among the geography units of that area frame. A value of 100 indicates that the geography unit has the highest community resilience among the geography units of that area frame. You can review more about the logical implications of the geometric mean in the 'Geometric Mean Discussion' section of this document.

Step 4 - Postprocess Index

At this stage, we have the Overall Theme, Socioeconomic Theme, and Housing Theme values across both geography units and all area frames. We exported this data from the python script to ArcGIS Pro. In this program, we join the Census tract and municipality data to their respective maps. For each map, we then classified the range of 0-100 theme values into a quantile classification, which created 4 classes with the same number of features per class. We classified theme values of 75 – 100 as "High Resilience". We classified theme values of 50 – 75 as "Medium-High Resilience". We classified theme values of 25 – 50 as "Medium-Low Resilience". We classified theme values of 0 – 25 as "Low Resilience".

Geometric Mean Discussion

The geometric mean is a non-compensatory index construction method. This means that each percentile rank value has a direct influence on the final index calculation. In other words, a high percentile value in one indicator does not compensate for a low percentile value in a different indicator. Conversely, a compensatory method such as measuring the sum of percentile values and then dividing by the number of summed percentile values would result in high and low indicator values to effectively cancel each other. There are two observed characteristics that non-compensatory methods have on the index value (Munda and Nardo, 2005):

1. A geography unit where all indicators are ranked around the middle could perform better than a geography unit with low extreme indicator values, even if the arithmetic mean of indicators for the latter geography unit is higher than the former.

2. An increase in low indicator values has a greater influence on the index value than an increase in high indicator values.

The implication of these two characteristics of non-compensatory methods such as the geometric mean are defensible in the topical context of the Community Resilience Index. For example, let's imagine two Census tracts. In Census tract A, the percentile value of all indicators is 50. In Census tract B, one indicator has a percentile value of 95 while the remaining indicators have a percentile value below 40. It's likely that a non-compensatory geometric mean method to calculate the composite index will have a higher index value for Census tract A. On the other hand, it's likely that a compensatory arithmetic mean method will result in a higher index value for Census tract B. Ultimately, a final composite index value should reflect the socioeconomic and housing conditions of the entire population, rather than favoring only the population represented by an indicator with a high percentile value. With this consideration, a non- compensatory geometric mean method better reflects the anticipated outcome of the CRI.

Indicator Displays

What is the Standard Mean Difference?

After the composite index, represented in the form of 3 different themes, the indicator displays are the second main component of the CRI. In the indicator displays component of the CRI, a user can analyze the spatial distribution of the 11 normalized indicator values used in the construction of the composite index. The values calculated from the statistic standard mean difference represents spatial distribution of normalized indicator values. This statistic standardizes the distance of each geography unit's normalized indicator value from the area frame's normalized indicator mean.

The standard mean difference is represented by an output at each geography unit for each indicator that ranges from -1 to 1. A geography unit with a standard mean difference of 0 indicates that the geography unit's normalized indicator value is equivalent to the normalized indicator mean of the selected area frame. A negative value from the standard mean difference indicates that the geography unit's normalized indicator value is below the area frame's normalized indicator mean. Conversely, a positive value indicates that the geography unit's normalized indicator value is above the area frame's normalized indicator mean. Furthermore, the greater the absolute value of the output, the farther that geography unit's normalized indicator value is from the area frame's mean. In other words, values near the extremes of either -1 or 1 indicate greater difference from the area frame's normalized mean than values closer to 0. The display of these values on a map enables a user to quickly identify the distribution of an indicator across an area frame.

Standard Mean Difference Calculation Steps

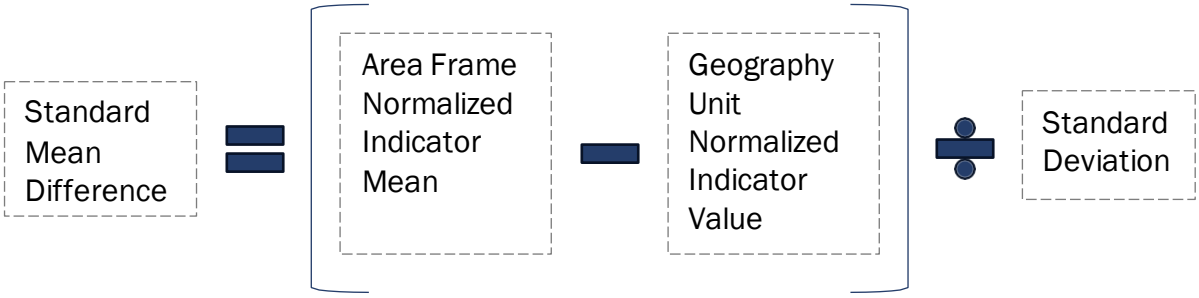


Figure 3. A visual representation of the standard mean difference equation.

Step 1 – Area Frame Indicator Means

At each area frame, determine the area frame's normalized indicator mean. This is completed in two different ways. For the statewide and individual county area frames, the normalized indicator mean is determined in the same manner as the normalized indicator values of specific Census tracts and municipalities. This means acquire the specific variables from the US Census Bureau ACS data tables necessary to construct each indicator, and then process those variables in a python script to create the normalized indicator. The 'Index Construction Steps' section of this document outlines this processing step in more detail.

The normalized indicator means for the area frames of VDH districts and hospital service areas require a different approach since ACS data tables do not directly provide data for these geographies. However, the boundaries of these area frames are determined by the boundaries of municipalities. This means that each municipality belongs to a single VDH district and hospital service area. As such, to determine the normalized indicator average for VDH Districts and hospital service areas, take the normalized indicator average of the municipalities that belong to each of these area frames.

Step 2 –Difference Mean

At each geography unit, subtract the normalized indicator value from the area frame's normalized indicator mean. This results in the normalized difference between the area frame mean and the geography unit's value.

Step 3 – Standard Deviation

Calculate the standard deviation of the normalized indicator values of all geography units for each area frame.

Step 4 – Standard Mean Difference

Divide the mean difference by the standard deviation to determine the standard mean difference of each geography unit's normalized indicator value compared to the area frame's standardized indicator mean.

Disaggregated Demographics

Disaggregated demographic data of indicators is a second aspect of the indicator displays component of the CRI. The demographic data is disaggregated by race and ethnicity groups as defined by the US Census Bureau. Due to limited data availability across the ACS variables, only 8 out of the 11 indicators provide disaggregated data. In all cases, the provided scale of disaggregated data is an individual geography unit, but rather the entire area frame. That means when a user reviews the data for a geography unit, the disaggregated demographic data is the mean for the area frame and is not the data of that specific geography unit. The presentation of disaggregated data as the area frame's mean accommodates the small population values present among some demographic groups at the individual geography unit. Refer to the 'Data Appendix' section of the document to review the ACS variables used to construct the disaggregated indicator values.

Data Appendix

Download the table of ACS variables and associated metadata used to construct CRI indicators.

Download the table of ACS variables and associated metadata used to construct CRI indicators disaggregated by demographic groups.

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